



Assessment I - Skills Integration Challenge I

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13.1 Objective

The Objectives of this lab are:

- Configure basic router security
- Configure basic switch security
- Configure AAA local authentication
- Configure SSH
- Secure against login attacks
- Configure site-to-site IPsec VPNs
- Configure firewall and IPS settings
- Configure ASA basic security and firewall settings

13.2 Background/Scenario

This culminating challenge includes many of the skills that you have acquired during this course. The routers and switches are preconfigured with the basic device settings, such as IP addressing and routing. You will secure routers using the CLI to configure various IOS features, including AAA, SSH, and Zone-Based Policy Firewall (ZPF). You will configure a site-to-site VPN between R1 and R3. You will secure the switches on the network. In addition, you will also configure firewall functionality on the ASA. At the end of the challenge, attach your scripts for R1, R2, R3, S1, S2 and ASA.

13.3 Topology

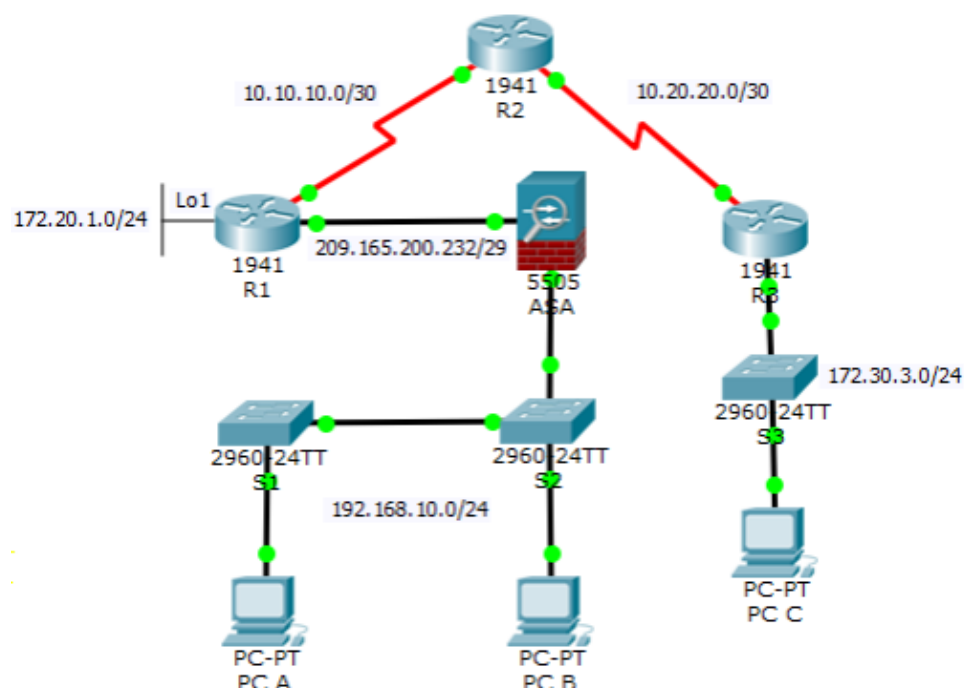




Figure 1: Topology

13.4 Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	209.165.200.233	255.255.255.248	N/A
	S0/0/0 (DCE)	10.10.10.1	255.255.255.252	N/A
	Loopback 1	172.20.1.1	255.255.255.0	N/A
R2	S0/0/0	10.10.10.2	255.255.255.252	N/A
	S0/0/1 (DCE)	10.20.20.2	255.255.255.252	N/A
R3	G0/1	172.30.3.1	255.255.255.0	N/A
	S0/0/1	10.20.20.1	255.255.255.252	N/A
S1	VLAN 1	192.168.10.11	255.255.255.0	192.168.10.1
S2	VLAN 1	192.168.10.12	255.255.255.0	192.168.10.1
S3	VLAN 1	172.30.3.11	255.255.255.0	172.30.3.1
ASA	VLAN 1 (E0/1)	192.168.10.1	255.255.255.0	N/A
	VLAN 2 (E0/0)	209.165.200.234	255.255.255.248	N/A
PC-A	NIC	192.168.10.2	255.255.255.0	192.168.10.1
PC-B	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PC-C	NIC	172.30.3.3	255.255.255.0	172.30.3.1

Figure 2: Addressing Table

Task 1: Configure Basic Router Settings

- Configure the following on R1:
 - Minimum password length is 10 characters.
 - Encrypt plaintext passwords.
 - Privileged EXEC mode secret password is **ciscoenapa55**.
 - Console line password is **ciscoconpa55**, timeout is **15** minutes, and console messages should not interrupt command entry.
 - A message-of-the-day (MOTD) banner should include the word **unauthorized**.



```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#security passwords min-length 10
R1(config)#service password-encryption
R1(config)#enable secret ciscoenpa55
R1(config)#line con 0
R1(config-line)#password ciscoconpa55
R1(config-line)#exec-timeout 15 0
R1(config-line)#logging synchronous
R1(config-line)#login
R1(config-line)#exit
R1(config)#banner motd # unauthorized access is strictly prohibited #
R1(config)#
```

- Configure the following on R2:
 - Privileged EXEC mode secret password is **ciscoenapa55**.
 - Password for the VTY lines is **ciscovtypa55**, timeout is **15** minutes, and login is required.

```
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#enable secret ciscoenpa55
R2(config)#line vty 0 4
R2(config-line)#password ciscovtypa55
R2(config-line)#exec-timeout 15 0
R2(config-line)#login
R2(config-line)#exit
R2(config)#
```

Task 2: Configure Basic Switch Security

- Configure the following on S1:
 - Encrypt plaintext passwords.
 - Privileged EXEC mode secret password is **ciscoenapa55**.
 - Console line password is **ciscoconpa55**, timeout is **5** minutes, and consoles messages should not interrupt command entry.
 - Password for the VTY lines is **ciscovtypa55**, timeout is **5** minutes, and login is required.
 - An MOTD banner should include the word **unauthorized**



```
S1>en
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#service password-encryption
S1(config)#enable secret ciscoenpa55
S1(config)#line con 0
S1(config-line)#password ciscoconpa55
S1(config-line)#exec-timeout 5 0
S1(config-line)#logging synchronous
S1(config-line)#login
S1(config-line)#exit
S1(config)#line vty 0 15
S1(config-line)#password ciscovtypa55
S1(config-line)#exec-timeout 5 0
S1(config-line)#login
S1(config-line)#exit
S1(config)#banner motd # unauthorized access is strictly prohibited #
```

- Configure trunking between S1 and S2 with the following settings:
 - Set the mode to trunk and assign VLAN 99 as the native VLAN.
 - Disable the generation of DTP frames.

```
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#int g0/1
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk native vlan 99
S1(config-if)#switchport nonegotiate
S1(config-if)#exit
```

- Configure the S1 with the following port settings:
 - F0/6 should only allow access mode, set to **PortFast**, and enable BPDU guard.
 - F0/6 uses basic default port security with dynamically learned MAC addresses added to the running configuration.
 - All other ports should be disabled

```
S1(config)#int f0/6
S1(config-if)#switchport mode access
S1(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/6 but will only
have effect when the interface is in a non-trunking mode.
S1(config-if)#spanning-tree bpduguard enable
S1(config-if)#switchport port-security
S1(config-if)#switchport port-security mac-address sticky
S1(config-if)#exit
S1(config)#int range f0/2-5, f0/7-24, g0/2
S1(config-if-range)#shutdown
```

Task 3: Configure AAA Local Authentication

- Configure the following on R1:



- Create a local user account of **Admin01**, a secret password of **Admin01pa55**, and a privilege level of **15**.
- Enable AAA services.
- Implement AAA services using the local database as the first option and then the **enable** password as the backup option.

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#username Admin01 privilege 15 secret Admin01pa55
R1(config)#aaa new-model
R1(config)#aaa authentication login default local enable
R1(config)#
```

Task 4: Configure SSH

- Configure the following on R1:
 - The domain name is **ccnasecurity.com**
 - The RSA key should be generated with **1024** modulus bits.
 - Only SSH version 2 is allowed.
 - Only SSH is allowed on VTY lines.

```
R1(config)#ip domain-name ccnasecurity.com
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
```

```
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

```
R1(config)#ip ssh version 2
*Mar 1 1:7:8.419: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#
```

- Verify that PC-C can remotely access R1 (209.165.200.233) using SSH.

Attach screenshot below:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l Admin01 209.165.200.233

Password:
R1>
```

Task 5: Secure Against Login Attacks

- Configure the following on R1:
 - If a user fails to log in twice within a 30-second time span, disable logins for one minute.



- Log all failed login attempts.

```
R1(config)#login block-for 60 attempts 2 within 30
R1(config)#login on-failure log
R1(config)#
```

Task 6: Configure Site-to-Site IPsec VPNs

- Enable the Security Technology package license on R1
 - Save the running configuration before reloading
- Configure the following on R1:
 - Create an access list to identify interesting traffic on R1.
 - Configure ACL **101** to allow traffic from the R1 Lo1 network to the R3 G0/1 LAN
- Configure the **crypto isakmp policy 10** Phase 1 properties on R1 and the shared crypto key **ciscovnpa55**. Use the following parameters:
 - Key distribution method: **ISAKMP**
 - Encryption: **aes 256**
 - Hash: **sha**
 - Authentication method: **pre-shared**
 - Key exchange: **DH Group 5**
 - IKE SA lifetime: **3600**
 - ISAKMP key: **ciscovnpa55**
- Create the transform set **VPN-SET** to use **esp-aes 256** and **esp-sha-hmac**. Then create the crypto map **CMAPI** that binds all of the Phase 2 parameters together. Use sequence number **10** and identify it as an **ipsec-isakmp** map. Use the following parameters:
 - Transform set: **VPN-SET**
 - Transform encryption: **esp-aes 256**
 - Transform authentication: **esp-sha-hmac**
 - Perfect Forward Secrecy (PFS): **group5**
 - Crypto map name: **CMAPI**
 - SA establishment: **ipsec-isakmp**
 - Bind the crypto map (**CMAPI**) to the outgoing interface.



```
Username: Admin01
Password:
R1>en
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 101 permit ip 172.20.1.0 0.0.0.255 172.30.3.0 0.0.0.255
R1(config)#
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#hash sha
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#lifetime 3600
R1(config-isakmp)#exit
R1(config)#crypto isakmp key ciscovpnpa55 address 10.20.20.1
R1(config)#crypto ipsec transform-set VPN-set esp-aes 256 esp-sha-hmac
R1(config)#crypto map CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
      and a valid access list have been configured.
R1(config-crypto-map)#set transform-set VPN-set
R1(config-crypto-map)#set pfs group 5
% Ambiguous command: "set pfs group 5"
R1(config-crypto-map)#set pfs group5
R1(config-crypto-map)#match address 101
R1(config-crypto-map)#set peer 10.20.20.1
R1(config-crypto-map)#exit
R1(config)#int s0/0/0
R1(config-if)#crypto map CMAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
R1(config-if)#exit
R1(config)#
```

- Verify that the Security Technology package license is enabled. Repeat the site-to-site VPN configurations on R3 so that they mirror all configurations from R1.



```
R3>en
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#access-list 101 permit ip 172.30.3.0 0.0.0.255 172.20.1.0 0.0.0.255
R3(config)#crypto isakmp policy 10
R3(config-isakmp)#encryption aes 256
R3(config-isakmp)#hash sha
R3(config-isakmp)# authentication pre-share
R3(config-isakmp)#group 5
R3(config-isakmp)#lifetime 3600
R3(config-isakmp)#exit
R3(config)#crypto isakmp key ciscovpnpa55 address 10.10.10.1
R3(config)#crypto ipsec transform-set VPN-set esp-aes 256 esp-sha-hmac
R3(config)#crypto map CMAP 10 ipsec-isakmp
% Invalid input detected at '^' marker.
```

```
R3(config)#crypto ipsec transform-set VPN-set esp-aes 256 esp-sha-hmac
R3(config)#crypto map CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
R3(config-crypto-map)#set transform-set VPN-SET
ERROR: transform set with tag VPN-SET does not exist.
R3(config-crypto-map)#set transform-set VPN-set
R3(config-crypto-map)#set pfs group5
R3(config-crypto-map)#match address 101
R3(config-crypto-map)#set peer 10.10.10.1
R3(config-crypto-map)#exit
R3(config)#int s0/0/1
R3(config-if)#crypto map CMAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
R3(config-if)#exit
R3(config)#
```

- Ping the Lo1 interface (172.20.1.1) on R1 from PC-C. On R3, use the **show crypto ipsec sa** command to verify that the number of packets is more than 0, which indicates that the IPsec VPN tunnel is working

```
Pinging 172.20.1.1 with 32 bytes of data:

Request timed out.
Reply from 172.20.1.1: bytes=32 time=2ms TTL=254
Reply from 172.20.1.1: bytes=32 time=2ms TTL=254
Reply from 172.20.1.1: bytes=32 time=2ms TTL=254

Ping statistics for 172.20.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>
```




```
R3#show crypto ipsec sa

interface: Serial0/0/1
  Crypto map tag: CMAP, local addr 10.20.20.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (172.30.3.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (172.20.1.0/255.255.255.0/0/0)
current_peer 10.10.10.1 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
#pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.20.20.1, remote crypto endpt.:10.10.10.1
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/1
current outbound spi: 0x705EE20B(1885266443)

inbound esp sas:
  spi: 0xE5F82B14(3858246420)
    transform: esp-aes 256 esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2003, flow_id: FPGA:1, crypto map: CMAP
    sa timing: remaining key lifetime (k/sec): (4525504/3546)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0x705EE20B(1885266443)
    transform: esp-aes 256 esp-sha-hmac ,
    in use settings ={Tunnel, }
    conn id: 2004, flow_id: FPGA:1, crypto map: CMAP
    sa timing: remaining key lifetime (k/sec): (4525504/3546)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

outbound ah sas:

outbound pcp sas:
```

Task 7: Configure Firewall and IPS Settings

- Configure a ZPF on R3 using the following requirements:
 - Create zones named **IN-ZONE** and **OUT-ZONE**.
 - Create an ACL number **110** that defines internal traffic, which permits all IP protocols from the **172.30.3.0/24** source network to **any** destination.



- Create a class map named **INTERNAL-CLASS-MAP** that uses the **match-all** option and **ACL 110**
- Create a policy map named **IN-2-OUT-PMAP** that uses the class map **INTERNAL-CLASS-MAP** to **inspect** all matched traffic.
- Create a zone pair named **IN-2-OUT-ZPAIR** that identifies **IN-ZONE** as the source zone and **OUT-ZONE** as the destination zone.
 - Specify that the **IN-2-OUT-PMAP** policy map is to be used to **inspect** traffic between the two zones.
 - Assign **G0/1** as an **IN-ZONE** member and **S0/0/1** as an **OUT-ZONE** member.

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#zone security IN-ZONE
R3(config-sec-zone)#zone security OUT-ZONE
R3(config-sec-zone)#exit
R3(config)#zone security OUT-ZONE
R3(config-sec-zone)#exit
R3(config)#access-list 110 permit ip 173.30.3.0 0.0.0.255 any
R3(config)#class-map type inspect match-all INTERNAL-CLASS-MAP
R3(config-cmap)#match access-group 110
R3(config-cmap)#exit
R3(config)#policy-map type inspect IN-2-OUT-PMAP
R3(config-pmap)#class type inspect INTERNAL-CLASS-MAP
R3(config-pmap-c)#inspect
%No specific protocol configured in class INTERNAL-CLASS-MAP for inspection. All
protocols will be inspected
R3(config-pmap-c)#exit
R3(config-pmap)#exit
R3(config)#zone-pair security IN-2-OUT-ZPAIR source IN-ZONE destination OUT-ZONE
R3(config-sec-zone-pair)#service-policy type inspect IN-2-OUT-PMAP
R3(config-sec-zone-pair)#exit
R3(config)#int g0/1
R3(config-if)#zone-member security IN-ZONE
R3(config-if)#exit
R3(config)#int s0/0/1
R3(config-if)#zone-member security OUT-ZONE
R3(config-if)#exit
R3(config)#
```

- Configure an IPS on R3 using the following requirements:

Note: Within Packet Tracer, the routers already have the signature files imported and in place. They are the default XML files in flash. For this reason, it is not necessary to configure the public crypto key and complete a manual import of the signature files.

- Create a directory in flash named **ipsdir** and set it as the location for IPS signature storage.
- Create an IPS rule named **IPS-RULE**.
- Retire the **all** signature category with the **retired true** command (all signatures within the signature release).
- Unretire the **IOS_IPS Basic** category with the **retired false** command.
- Apply the rule inbound on the **S0/0/1** interface



```
R3#mkdir ipsdir
Create directory filename [ipsdir]?
Created dir flash:ipsdir

R3#ip ips name IPS-RULE
^
% Invalid input detected at '^' marker.

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip ips name IPS-RULE
R3(config)#ip ips config location flash:ipsdir
R3(config)#ip ips signature-category
R3(config-ips-category)#category all
R3(config-ips-category-action)#retired true
R3(config-ips-category-action)#exi
R3(config-ips-category)#category ios_ips basic
R3(config-ips-category-action)#retired false
R3(config-ips-category-action)#exit
R3(config-ips-category)#exit
Do you want to accept these changes? [confirm]yes

R3(config)#int s0/0/1
R3(config-if)#ip ips IPS-RULE in
```

Task 8: Configure ASA Basic Security and Firewall Settings

- Configure VLAN interfaces with the following settings:
 - For the VLAN 1 interface, configure the addressing to use **192.168.10.1/24**.
 - For the VLAN 2 interface, remove the default DHCP setting and configure the addressing to use **209.165.200.234/29**.
- Configure hostname, domain name, enable password, and console password using the following settings:
 - The ASA hostname is **CCNAS-ASA**.
 - The domain name is **ccnasecurity.com**.
 - The enable mode password is **ciscoenapa55**.
- Create a user and configure AAA to use the local database for remote authentication.
 - Configure a local user account named **admin** with the password **adminpa55**. Do not use the **encrypted** attribute.
 - Configure AAA to use the local ASA database for SSH user authentication.
 - Allow SSH access from the outside host **172.30.3.3** with a timeout of **10** minutes
- Configure the ASA as a DHCP server using the following settings:
 - Assign IP addresses to inside DHCP clients from 192.168.10.5 to 192.168.10.30.
 - Enable DHCP to listen for DHCP client requests.
- Configure static routing and NAT.
 - Create a static default route to the next hop router (R1) IP address.
 - Create a network object named **inside-net** and assign attributes to it using the **subnet** and **nat** commands.
 - Create a dynamic NAT translation to the outside interface.



- Modify the Cisco Modular Policy Framework (MPF) on the ASA using the following settings:
 - Configure **class-map inspection_default** to **match default-inspection-traffic**, and then exit to global configuration mode.
 - Configure the **policy-map** list **global_policy**. Enter the **class inspection_default** and enter the command to **inspect icmp**. Then exit to global config mode.
 - Configure the MPF **service-policy** to make the **global_policy** apply globally.

```
ciscoasa#conf t
ciscoasa(config)#int vlan 1
ciscoasa(config-if)#ip address 192.168.10.1 255.255.255.0
Waiting for the earlier webvpn instance to terminate...
Previous instance shut down.Starting a new one
ciscoasa(config-if)#exit
ciscoasa(config)#int vlan 2
ciscoasa(config-if)#no ip address dhcp
WARNING: DHCPD bindings cleared on interface 'outside', address pool removed
ciscoasa(config-if)#ip address 209.165.200.234 255.255.255.248
Waiting for the earlier webvpn instance to terminate...
Previous instance shut down.Starting a new one
ciscoasa(config-if)#exit
ciscoasa(config)#hostname CCNAS-ASA
CCNAS-ASA(config)#domain-name ccnasecurity.com
CCNAS-ASA(config)#enable password ciscoenapa55
CCNAS-ASA(config)#username admin password adminpa55
CCNAS-ASA(config)#aaa authentication ssh console local
CCNAS-ASA(config)#crypto key generate rsa modulus 1024
WARNING: You have a RSA keypair already defined named <Default-RSA-Key>.

Do you really want to replace them? [yes/no]: no
ERROR: Failed to create new RSA keys named <Default-RSA-Key>
CCNAS-ASA(config)#ssh 172.30.3.3 255.255.255.255 outside
CCNAS-ASA(config)#ssh timeout 10
CCNAS-ASA(config)#dhcpd address 192.168.10.5-192.168.10.30 inside
CCNAS-ASA(config)#dhcpd enable inside
need to define address pool range first
dhcpd enable command failed
CCNAS-ASA(config)#route outside 0.0.0.0 0.0.0.0 209.165.200.233
CCNAS-ASA(config)#object network inside-net
CCNAS-ASA(config-network-object)#subnet 192.168.10.0 255.255.255.0
CCNAS-ASA(config-network-object)#nat (inside,outside) dynamic interface
CCNAS-ASA(config-network-object)#exit
CCNAS-ASA#conf t
CCNAS-ASA(config)#class-map inspection_default
CCNAS-ASA(config-cmap)#match default-inspection-traffic
CCNAS-ASA(config-cmap)#exit
CCNAS-ASA(config)#policy-map global_policy
CCNAS-ASA(config-pmap)#class inspection_default
CCNAS-ASA(config-pmap-c)#inspect icmp
      ^
% Invalid input detected at '^' marker.

CCNAS-ASA(config-pmap-c)#inspect icmp
CCNAS-ASA(config-pmap-c)#exit
CCNAS-ASA(config)#service-policy global_policy global
CCNAS-ASA(config)#
```



```
C:\>ping 209.165.200.233

Pinging 209.165.200.233 with 32 bytes of data:

Request timed out.
Reply from 209.165.200.233: bytes=32 time<1ms TTL=254
Reply from 209.165.200.233: bytes=32 time<1ms TTL=254
Reply from 209.165.200.233: bytes=32 time<1ms TTL=254

Ping statistics for 209.165.200.233:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```



Attach scripts for R1, R2, R3, S1, S2 and ASA below.

Congratulations Guest! You completed the activity.

Overall Feedback

Assessment Items

Connectivity Tests

Expand/Collapse All

Show Incorrect Items

Assessment Items	Status	Points	Component(s)	Feedt
- Network				
- ASA				
- AAA		0	Other	
- Authentication		0	Other	
- Authentication SSH	Correct	1	Other	
- ACL		0	ACL	
- ?acl_inside-net?	Correct	1	ACL	
- Class Maps		0	Other	

Score : 94/94

Item Count : 94/94

Component	Items/Total	Score
ACL	5/5	5/5
Ip	24/24	24/24
Other	53/53	53/53
Physical	4/4	4/4
Switching	8/8	8/8



Assessment Rubric
Assessment I - Skills Integration Challenge I

Name:	Student ID:
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Points Distribution

Task No.	LR 2 Simulation	LR5 Results/Plots	LR9 Report
Task 1	5		
Task 2	15		
Task 3	5		
Task 4	5	5	
Task 5	5	5	
Task 6	15		
Task 7	15		
Task 8	15		
Total	80	/10	/10
CLO Mapped	CLO 1-4	CLO 1-4	CLO 1-4

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission	/10	CLO 4

CLO	Total Points	Points Obtained
1-4	90	
1-4	10	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR2	Program/Code / Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR9	Report	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.